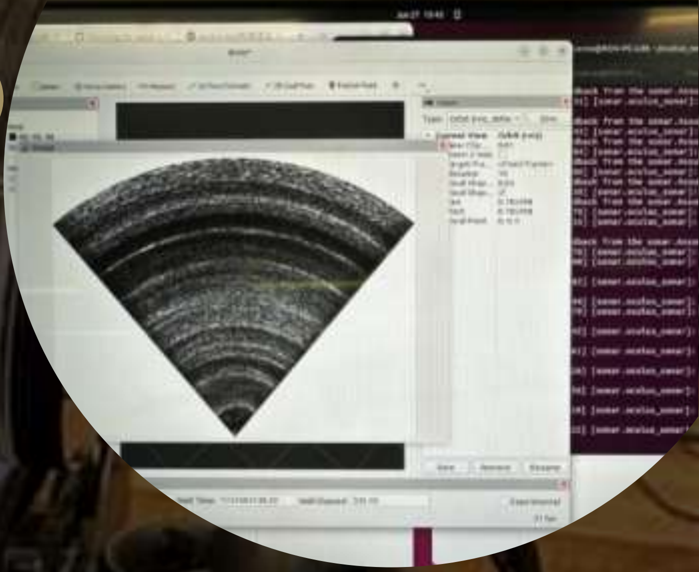
Projects Portfolio

Kushagra Javeri



Research assistant – Underwater Pylon inspection

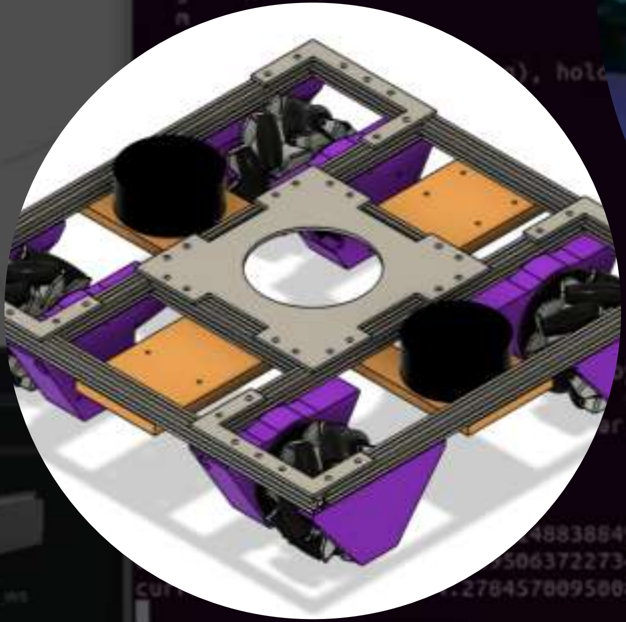
- Worked as a research assistant at UNSW for underwater robotics research in autonomous mapping and pylon inspection.
- Integrated a high-end sensor suite on the BlueROV2 Heavy for autonomous capabilities, including sonar, DVL and stereo camera.
- Ran several field tests in pools and harbors, to depths up to 12 m.
- Post processed sonar and stereo data into point clouds for underwater SLAM and 3D reconstruction.

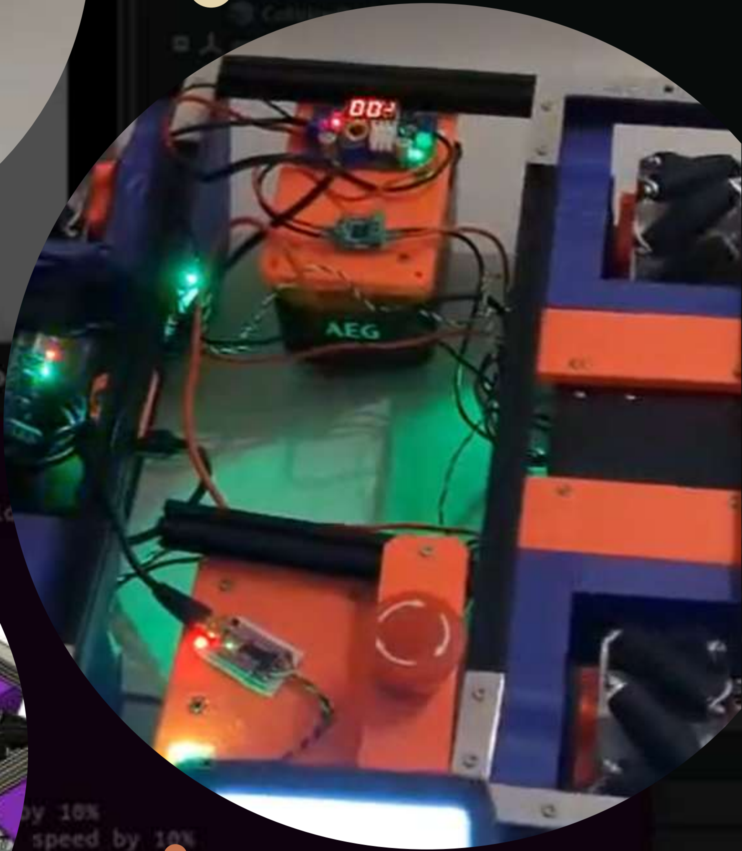
Singapore Autonomous Underwater Vehicle Challenge (SAUVC)

- Individually competed in SAUVC 2026, placing 2nd internationally. Developed a custom feature-based EKF-SLAM algorithm
- Team lead for Australia’s first competitive underwater robotics team
- Designed and assembled electronics, sensors and chassis for several AUVs.
- Developed ROS2 architecture for path planning, computer vision, mapping and navigation from scratch.
- Worked with Ardusub, ROS2, Mavlink protocol, YOLOv26n and BlueOS.
- 2nd place in the qualifiers in Singapore in SAUVC 2025

Robocup@Work 2026 – Undergraduate thesis and paper (Ongoing)

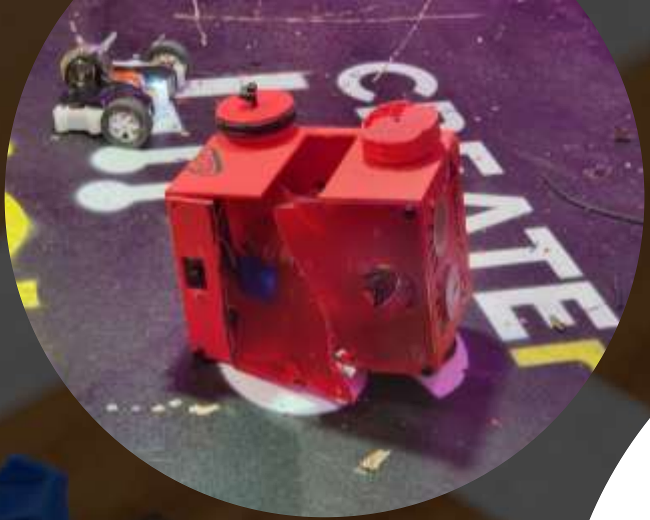
- Designed a custom build for an industrial holonomic robot, with a 6 DOF arm.
- Working on control algorithms for workstation alignment
- Reinforcement learning, PID, sliding mode controllers and model predictive control
- Used ROS2, Nav2 and slam_toolbox.
- Developed IsaacSim simulation with roller geometry.





Battlebots

- Made 150g robots that fight each other
- Participated and hosted multiple university-level battlebots competitions, first of its kind in Australia.
- Remote controlled robots with spinning blades and flippers (or both)
- Extensive use of Metal working, laser cutting, casting, CNC machining and 3D printing



DRC (Droid Racing Challenge) 2024

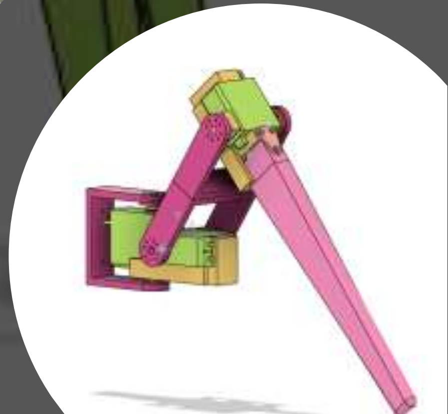
- Entered as a solo competitor representing UNSW
- Created my own new path planning algorithm inspired from artificial potential fields and magnetic fields
- Completely custom design, handmade wheels.
- Programmed on a Teensy and OpenMV Cam.
- Placed second in Australia. Mentored the winning team.

Contactile Pty Ltd (Engineering intern)

- Developed a first of its kind slip detection algorithm that detects and identifies translational and rotational incipient slip and provides an accurate friction estimate.
- Developed prototypes for dual axis sensor mounts and linear gripper arms.
- Worked on center of mass estimation algorithms using tactile sensor data, robotic arm and angular gripper manipulation.

SpiderBot

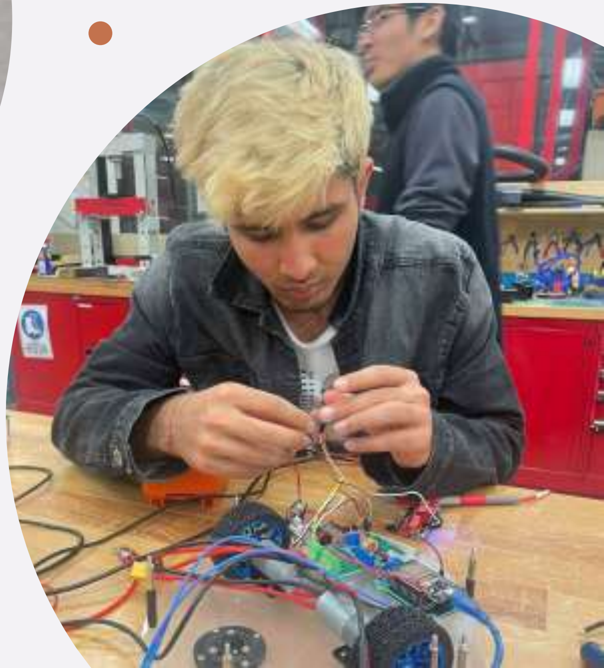
- Six legged remotely controlled robot with functionality for omnidirectional walking (tripod, wave, ripple) and turning on the spot.
- 3 servos in each leg for control, multi-terrain movement, climbing small steps and obstacle avoidance
- Extensive use of OOP and inverse kinematics to develop a codebase for gaits and control in C++.



DRC (DROID RACING Challenge) 2022

- Used ESP32 and OpenMV camera for blob detection and logic decisions
- Winners of Best droid as well as second in race
- Made design decisions regarding shape and structure of the droid for software to run smoothly.
- Modular design using laser cut acrylic plates for efficient testing and troubleshooting



More projects

- Worked on embedded systems teams with a TM4C.
- Introduced to ROS2 and Raspberry Pi when making a mars rover.
- Solar car: Laser cutting grooves in balsa wood to bend it securely.